



Dollars per Living Tree

Cost-effectiveness of kerbside watering-bag rounds against a canopy target that books only plantings

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Motivation

A metropolitan council bags its new street trees through their first two summers and reports canopy progress in trees planted. The watering round has never been priced against what it keeps alive. What does a round buy, per tree still standing?

Objectives

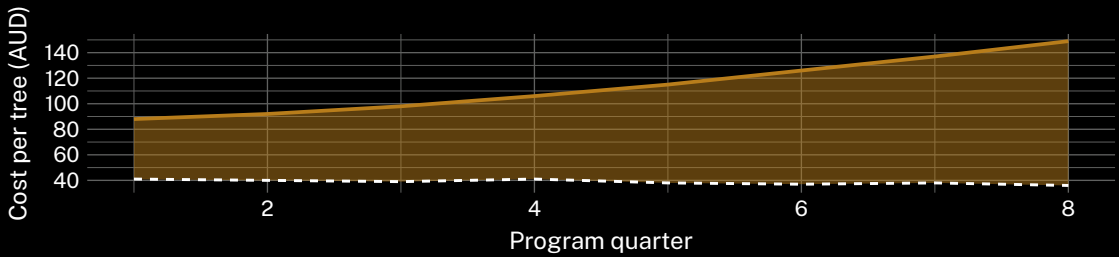
- Cost the watering-bag round per additional tree surviving to establishment.
- Test whether bagging dominates replacement planting at the council's own installed replacement cost.
- Trace what the canopy indicator does with the resulting figure.

Study design

- 386 street trees · 24 verge segments · two depots · 24 months spanning two summers.
- Matched-pair allocation by segment; rounds costed from depot timesheets, metered water, and bag amortisation; survival assessed at 24 months by an assessor blind to allocation.
- Cost parameters and the dominance threshold were pre-registered; none were adjusted post hoc.

Findings

Survival to establishment reaches 71.4% under bagged rounds against 46.9% unbagged, at an incremental \$318 per additional tree alive. Results suggest bagging dominates replacement planting wherever an installed replacement tree costs the council more than \$173.



Dashed: cost per tree planted, as the canopy target books it — \$41 to \$36 across the eight quarters to June 2026. Solid: cost per tree alive after two summers, rising \$88 to \$149.

Implications & takeaways

The program is cost-effective and the indicator cannot see it: a target denominated in plantings books the cheapest number the program produces. Costing the round per surviving tree moves that figure roughly fourfold and reorders which street the truck reaches first. Next: a survival-denominated variant of the canopy indicator, tabled with the Indicator Commons.

Selected references

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“Every bag we costed bought a tree the target could not count.”

71.4% of bagged trees reach establishment against 46.9% unbagged, at \$318 per additional tree alive.